

Food Insecurity Forecast — Model Review

Six models compared: three data-lag structures (current-year, 2-year, 3-year), each fit two ways (ridge and plain OLS). Trained on national Map the Meal Gap data, 2020–2024, evaluated specifically against Florida.

This page pulls together the coefficients and accuracy numbers for all six fitted models so we can walk through what each one is actually saying, side by side, rather than reading raw notebook output together.

How to read the years below. Every model predicts a **target year** — the actual food insecurity rate it's trying to match (Y) — using drivers and the prior rate measured in an earlier **source year** (X). For the lag models, source year = target year – lag. Example: the 3-year lag model trains on the **2020 → 2023** transition (2020 data predicting the real 2023 rate), then is tested on **2021 → 2024** (2021 data predicting the real 2024 rate) — a year it never saw during training.

Summary ALL SIX MODELS

Lower MAE / RMSE is better — both are measured in food insecurity rate (0–1 scale), population-weighted, and reported against Florida rows specifically since that's the actual target population.

MODEL	TRAIN		TEST		FLORIDA MAE	FLORIDA RMSE	
	Y	X	Y	X			
Current-year — ridge	2021–2023	same yr*	2024	same yr*	0.0160	0.0216	NEEDS SAME-YEAR DATA
Current-year — OLS	2021–2023	same yr*	2024	same yr*	0.0160	0.0216	NEEDS SAME-YEAR DATA
2-year lag — ridge	2022–2023	2020–2021	2024	2022	0.0276	0.0313	
2-year lag — OLS	2022–2023	2020–2021	2024	2022	0.0276	0.0313	
3-year lag — ridge	2023	2020	2024	2021	0.0179	0.0248	BEST USABLE MODEL
3-year lag — OLS	2023	2020	2024	2021	0.0237	0.0299	

Y = target year (the rate being predicted). **X** = source year (where drivers/prior rate are measured). *Current-year's drivers come from the same year as the target, but the lagged-rate feature is always 1 year before target, even in this spec — that's what "rate[t-1]" means. "Needs same-year data" = requires drivers from the same year being predicted, which FTB won't have on hand in practice given the data lag — kept here as an accuracy ceiling for comparison, not a candidate for real use.

Current-year drivers RATE[T] ~ RATE[T-1] + DRIVERS[T]

Predicts a target year's rate using that *same* year's economic and demographic drivers (source = target year), plus the prior year's rate as an added feature. Trained on target years 2021–2023 (each using its own year's drivers, and the year before as the lagged-rate input); tested on target year 2024 (2024 drivers, 2023 rate). Most accurate of the three by a small margin — but not realistic for FTB to run in practice, since it needs drivers from the same year being forecast, which won't exist yet at forecast time. Included here as the ceiling the lag models are measured against, not as a model to actually deploy.

Ridge and OLS land on effectively the same fit here (coefficients agree to 8 decimal places once un-standardized) — the regularization isn't doing anything at this sample size, so the plain, interpretable version costs nothing.

DRIVER	COEFFICIENT	WHAT IT MEANS
Last year's food insecurity rate	+0.8469	By far the strongest signal — a ZIP's own recent history.
County food cost index	−0.2102	County's food cost relative to the national average.
ALICE poverty household rate	+0.0902	FL-only; share of county households below the poverty line (ALICE definition).
Poverty rate	+0.0966	Share of ZIP residents below the poverty line.
Disability rate	+0.0717	Share of ZIP residents with a disability.
Unemployment rate	+0.0748	Local unemployment rate.
County cost per meal	+0.0527	Average dollar cost of a meal in the county.

DRIVER	COEFFICIENT	WHAT IT MEANS
ALICE financial insecurity rate	−0.0297	FL-only; share of county households below the ALICE survival threshold.
Child population share	−0.0265	County's share of residents who are children.
Homeownership rate	−0.0160	Share of homes owner-occupied.
Median income (log)	+0.0149	Log-scaled to reduce the influence of a few very high-income ZIPs.
Percent Hispanic	+0.0094	ZIP demographic composition.
County % white, non-Hispanic	+0.0057	County demographic composition.
County SNAP threshold	−0.0057	County's SNAP eligibility income threshold.
Percent Black	−0.0044	ZIP demographic composition.
Has ALICE data	−0.0011	Flag for whether this ZIP's county had ALICE data at all (Florida only).
Rural-urban code	−0.0004	USDA rural/urban classification.
ALICE threshold, under 65	≈0	FL-only; measured in raw dollars — tiny per \$1, not negligible over a \$10,000s range.
ALICE threshold, 65+	≈0	FL-only; same caveat as above.

Coefficients are in raw units: for the rate-type drivers (most of these), divide by 100 to read as "per 1 percentage point." Baseline (intercept): −0.166.

2-year lag $\text{RATE}[T] \sim \text{RATE}[T-2] + \text{DRIVERS}[T-2]$

Predicts a target year's rate using drivers and the prior rate both measured 2 years earlier (source = target − 2). Trained on the transitions **2020** → **2022** and **2021** → **2023** (source year → target year), pooled together; tested on **2022** → **2024** — 2022 data predicting the real 2024 rate. This is the first realistic option given FTB's data lag, but it comes out as the weakest of the three usable models here, and noticeably worse than current-year.

Ridge and OLS again agree almost exactly — no real difference, so read these as the interpretable version at no cost.

DRIVER	COEFFICIENT	WHAT IT MEANS
Rate 2 years earlier	+0.6372	Still the dominant signal, but weaker than at 1-year lag.
County food cost index (2y prior)	-0.1004	County's food cost relative to the national average, 2 years earlier.
Poverty rate (2y prior)	+0.0639	Share of ZIP residents below the poverty line, 2 years earlier.
Disability rate (2y prior)	+0.0603	Share of ZIP residents with a disability, 2 years earlier.
County cost per meal (2y prior)	+0.0296	Average dollar cost of a meal in the county, 2 years earlier.
Child population share (2y prior)	-0.0288	County's share of residents who are children.
ALICE poverty household rate (2y prior)	+0.0229	FL-only; share of county households below the poverty line (ALICE definition).
Homeownership rate (2y prior)	-0.0218	Share of homes owner-occupied.
Percent Hispanic (2y prior)	+0.0228	ZIP demographic composition.
Median income, log (2y prior)	-0.0136	Log-scaled median household income.
County SNAP threshold (2y prior)	-0.0072	County's SNAP eligibility income threshold.
Unemployment rate (2y prior)	+0.0329	Local unemployment rate, 2 years earlier.
Percent Black (2y prior)	-0.0041	ZIP demographic composition.
Has ALICE data	-0.0028	Flag for whether this ZIP's county had ALICE data at all (Florida only).
ALICE financial insecurity rate (2y prior)	-0.0052	FL-only; share of county households below the ALICE survival threshold.

DRIVER	COEFFICIENT	WHAT IT MEANS
County % white, non-Hispanic	≈ 0	County demographic composition.
Rural-urban code	$+0.0007$	USDA rural/urban classification.
ALICE threshold, under 65	≈ 0	FL-only; raw dollars — small per \$1, not negligible over a wide range.
ALICE threshold, 65+	≈ 0	FL-only; same caveat.

Baseline (intercept): 0.237.

3-year lag

RATE[T] ~ RATE[T-3] + DRIVERS[T-3]

RECOMMENDED

Predicts a target year's rate using drivers and the prior rate both measured 3 years earlier (source = target - 3) — the closest match to what FTB can realistically have on hand. Trained on the single transition **2020 → 2023** (2020 data predicting the real 2023 rate) — the only one available at this lag; tested on **2021 → 2024** (2021 data predicting the real 2024 rate, a year the model never trained on). It outperforms the 2-year lag model, which is why it's the practical pick despite reaching further back.

Ridge vs. OLS actually matters here — unlike the other two. This spec only had one usable training transition (2020 → 2023), the thinnest of the three. On that smaller, less varied training set, plain OLS overfits and comes out meaningfully worse (Florida MAE 0.0237 vs. ridge's 0.0179). **Use the ridge version** for this spec specifically — it's the one place the regularization is earning its keep.

	OLS	RIDGE
Florida MAE	0.0237	0.0179
Florida RMSE	0.0299	0.0248

Coefficients below reflect the pattern learned from the single 2020 → 2023 transition. They're the **OLS** values, shown for interpretability — ridge's coefficients are standardized internally and aren't in comparable "per 1-unit" terms. For actual scoring, use the persisted ridge model (better accuracy, per the callout above). All Florida-only ALICE coefficients, and the intercept itself, drop to exactly zero in this OLS fit — with just one training year and ALICE covering only

~2.5% of national rows, there wasn't enough signal left for an unregularized fit to use them; ridge keeps small nonzero values for the same terms instead.

DRIVER	COEFFICIENT	WHAT IT MEANS
Rate 3 years earlier	+0.6568	Still dominant — persistence in food insecurity holds up even 3 years out.
Disability rate (3y prior)	+0.0561	Share of ZIP residents with a disability, 3 years earlier.
Child population share (3y prior)	-0.0349	County's share of residents who are children.
Homeownership rate (3y prior)	-0.0315	Share of homes owner-occupied.
Poverty rate (3y prior)	+0.0365	Share of ZIP residents below the poverty line, 3 years earlier.
Percent Hispanic (3y prior)	+0.0308	ZIP demographic composition.
Unemployment rate (3y prior)	-0.0228	Local unemployment rate, 3 years earlier.
Median income, log (3y prior)	-0.0119	Log-scaled median household income.
Percent Black (3y prior)	+0.0089	ZIP demographic composition.
County SNAP threshold (3y prior)	-0.0074	County's SNAP eligibility income threshold.
County food cost index (3y prior)	-0.0075	County's food cost relative to the national average — same direction as the other two specs, just smaller.
County % white, non-Hispanic	≈0	County demographic composition.
Rural-urban code	+0.0010	USDA rural/urban classification.
County cost per meal (3y prior)	≈0	Average dollar cost of a meal in the county — nearly zero at this lag.
ALICE financial insecurity rate	0.0000	FL-only; dropped to zero — see note above.

DRIVER	COEFFICIENT	WHAT IT MEANS
ALICE poverty household rate	0.0000	FL-only; dropped to zero.
ALICE threshold, under 65	0.0000	FL-only; dropped to zero.
ALICE threshold, 65+	0.0000	FL-only; dropped to zero.
Has ALICE data	0.0000	FL-only flag; dropped to zero.

Baseline (intercept, OLS): 0.000 — collapsed to exactly zero, another symptom of fitting on a single training year. Ridge's intercept (0.149) is the one that's actually used at scoring time.

Recommendation

- **For actual deployment: the 3-year lag, ridge version.** It's the most accurate model among the two FTB can realistically run given the data lag, and it's the one spec where ridge's regularization is doing real, measurable work.
- **Current-year stays as a reference point only** — it's the most accurate overall, but needs data FTB won't have in time to be useful as a forecast.
- **2-year lag is the weakest usable option** and isn't a fallback we'd recommend over 3-year lag.
- **Revisit once 2025 data lands.** The 3-year lag model is trained on a single year (2023) — every additional year narrows the gap between ridge and OLS, and may eventually make the simpler, fully interpretable OLS version safe to use everywhere.

Source: Map the Meal Gap (national, 2020–2024) + Florida ALICE data. All six fitted models are saved as JSON under `models/cc/` in the project repo.